

Abstract

Multi-agent active inference gives a natural account of belief sharing: agents hold local posteriors over a shared latent state, communicate those beliefs, and pool them into a colony-level consensus. The same mechanism is fragile when a member is miscalibrated, corrupted, or strategically wrong. Because the standard pool multiplies the reports together, a single confident-but-wrong broadcast that puts near-zero mass on the true state can pull the whole consensus off it, outweighing many honest members. The colony therefore needs a way to preserve the useful structure of belief sharing while limiting the influence of contaminated beliefs.

Abstract I

This paper presents Active Fedference, a discrete-categorical framework that connects robust federated generalized variational inference with active inference belief sharing. The main bridge is structural: standard belief sharing appears as the non-robust corner of a broader generalized-Bayes family, while robust losses, conservative server fusion, and explicit aggregation diagnostics describe how the system moves away from that corner under declared contamination mechanisms. The result is not a replacement for belief sharing, but a containment result: ordinary belief sharing is recovered when robustness is turned off. Bounded-loss theory applies on the client axis, while the variational-server axis supplies an objective-backed redescending weight update.

The manuscript separates three robustness axes that are often conflated. First, client-side generalized-Bayes updates change how each agent absorbs evidence; this is the rigorous axis, carrying FedGVI's bounded-influence result only under the source theorem's loss, model, and contamination assumptions. Second, a sharp server-side reweighting heuristic suppresses beliefs that pull away from the emerging consensus, while carrying only its recovery-limit guarantee — no proven objective and no bounded-influence bound. Third, a variational aggregation rule supplies a more conservative objective-backed server alternative, with a raw effective-weight bound but not an estimator-level bounded-influence proof for the normalized consensus. Keeping these axes separate lets the paper state exactly which claims are proven, which are empirical, and which remain engineering extensions.

The study suite then exercises the framework as an end-to-end research system: recovery checks anchor the standard-Bayes limit, belief-sharing studies verify the communication baseline, contamination experiments test robust consensus, and extension studies probe moving agents, hierarchical latent structure, sensitivity to acuity and colony size, parameter recovery, and single-host socket-backed federation traces. All reported quantities are generated from deterministic analysis artifacts and injected into the manuscript by token, so the paper, figures, release package, and validation reports remain tied to the same execution record.

The open-source repository is `ActiveInferenceInstitute/Active_Fedference`. The production Zenodo release DOI is [10.5281/zenodo.21864004](https://doi.org/10.5281/zenodo.21864004), and the repository and deposited PDF point to each other through this DOI and the repository URL.

Keywords: active inference, federated learning, generalised variational inference, belief sharing, robustness, FedGVI

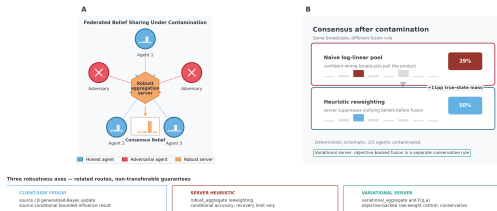
The complete system schematic is shown in fig. 1.

Abstract I

Federated belief sharing: from generative model to robust consensus

A categorical, recovery-biased bridge between active inference and generalized Bayes

(Recovery anchor: $\text{robust_aggregate}() = \text{log_linear_pool} \Rightarrow \text{standard belief sharing}$)



belief broadcasts $\Rightarrow$ fusion rule $\Rightarrow$ claim-bounded consensus schematic

Figure 1: Graphical abstract. Source relation: original project schematic; estimand: component relationships and recovery boundaries; uncertainty: none. **Recovery ribbon**: the zero-robustness identity anchors the construction at the standard log-linear belief-sharing pool. **Network panel**: the federated diagram shows 5 agents (3 honest, 2 adversarial) transmitting categorical beliefs to a central server. x-axis is agent position in the ring layout (left to right); y-axis/rows: each per-agent mini-bar glyph indexes posterior probability mass over hidden states. **Consensus panel**: deterministic outcome cards under 40% adversarial contamination compare the naive pool with canonical robust_aggregate heuristic reweighting; the displayed 39% and 50% are computed from the schematic beliefs. **Axis strip**: client-side FedGVI, server-side heuristic, and variational-server claims are shown as separate routes with non-transferable guarantees. This deterministic formal/mechanistic schematic has no CI, error band, or significance marker; it does not assign the variational server's objective-backed property to robust_aggregate.